

ICU LSTM Model – Test Case Results & Diet Recommendations

TestCase_ID	Category	Description	Expected_Behavior	Probability	Prediction	Validation_Status	Diet_Plan
1	Positive	Stable normal patient	Normal prediction, no alerts	0.422	Short Stay	OK	Balanced diet with fruits, vegetables, proteins
2	Positive	ICU patient with controlled vasopressors	Stable with intervention awareness	0.414	Short Stay	OK	Balanced diet with fruits, vegetables, proteins
3	Negative	Extremely high heart rate	Reject or clip HR	N/A	REJECTED	Rejected: invalid HR	N/A (invalid input)
4	Negative	Negative vital values	Input validation failure	N/A	REJECTED	Rejected: invalid HR	N/A (invalid input)
5	Negative	NaN values in labs	Graceful NaN handling	N/A	REJECTED	Rejected: unresolved NaN values	N/A (invalid input)
6	Negative	Medication logic conflict	Flag clinical inconsistency	0.418	Short Stay	OK	Balanced diet with fruits, vegetables, proteins
7	Negative	Negative medication values	Reject invalid meds	N/A	REJECTED	Rejected: negative Vasopressors	N/A (invalid input)
8	Neutral	Upper physiological boundary	Acceptable but cautious output	0.463	Short Stay	OK	Low glycemic index foods (oats, whole grains); Avoid sugar, sweets, sweetened drinks; Easily digestible foods (rice, soup); Avoid spicy and heavy meals
9	Neutral	Time-series HR jump	Handles anomaly smoothly	N/A	REJECTED	Rejected: invalid HR	N/A (invalid input)
10	Stress	Multi-organ failure scenario	High-risk output	0.4	Short Stay	OK	Low glycemic index foods (oats, whole grains); Avoid sugar, sweets, sweetened drinks; Low protein diet (avoid red meat); Restrict sodium and potassium; Easily digestible foods (rice, soup); Avoid spicy and heavy meals